Census Income Analysis

Classification & Customer Segmentation — Project Report

Prepared for: Retail Business Client | April 2026

1. Executive Summary

This report presents two machine learning models developed to support income-based marketing decisions for the retail client. The first model classifies individuals as earning above or below $50,000 annually. The second model segments the population into distinct customer groups for more targeted campaign planning. Both models were trained and evaluated on 1994-1995 U.S. Census Bureau data comprising 199,523 records and 40 demographic and employment variables.

The classification model produced two deployment-ready options:

- Tuned LightGBM (ROC-AUC 0.9374, Recall 0.86): Identifies 86% of all high earners. Recommended for campaigns where maximum reach is the priority.
- Ensemble Model with Threshold Tuning (ROC-AUC 0.9359, Precision 0.61, F1 0.62): More selective targeting. Of every 100 people flagged, 61 are actual high earners. Recommended when marketing efficiency matters as much as reach.

The segmentation model identified five distinct groups in the general adult population and two clearly differentiated groups within the high earner subset, each requiring a different marketing approach.

2. Business Objective

The retail client requires a reliable method to classify and understand potential customers by income level in order to direct marketing resources more effectively. Two specific needs were identified:

- Income classification: Given 40 demographic and employment variables, predict whether an individual earns above or below $50,000 annually. This enables the client to target high earners with premium product campaigns.
- Customer segmentation: Identify natural groupings within the population that go beyond the binary income split. Two people in the same income bracket may respond very differently to marketing based on their age, occupation, employment status, and financial behavior.

A key business judgment that informed the classification model: failing to identify a high earner (a missed opportunity) is more costly than incorrectly flagging a low earner (a wasted contact). This shaped the choice of evaluation metrics, with Recall on the high-income class treated as the primary performance indicator.

3. Data Overview

3.1 Dataset Summary

Property	Value
Source	U.S. Census Bureau, Current Population Surveys
Years Covered	1994 and 1995

Property	Value
Total Records	199,523
Features Available	40 demographic and employment variables
Target Variable	Annual income: above or below $50,000
Records Used for Modeling	143,531 (after removing minors)
Class Distribution (after cleaning)	91.4% under $50K / 8.6% over $50K

3.2 Key Findings from Data Exploration

Before building any models, we explored the data thoroughly to understand what variables drive income. Several patterns emerged clearly.

Age and income

High earners are concentrated between ages 35 and 60, with a median age of around 45. The under-$50K group spans a much wider range, reflecting economically inactive adults at both ends of the age spectrum.

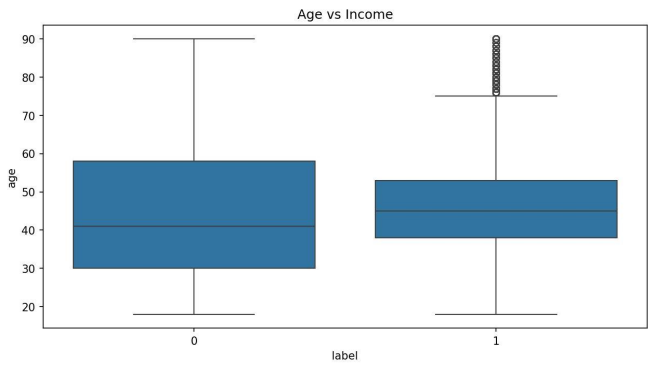

Figure 1: Age distribution by income group (0 = under $50K, 1 = over $50K)

Education and income

Education below Bachelor's degree level has near-zero representation in the high earner group. Almost all high earners hold at least a Bachelor's degree, though many degree holders still fall below the $50,000 threshold.

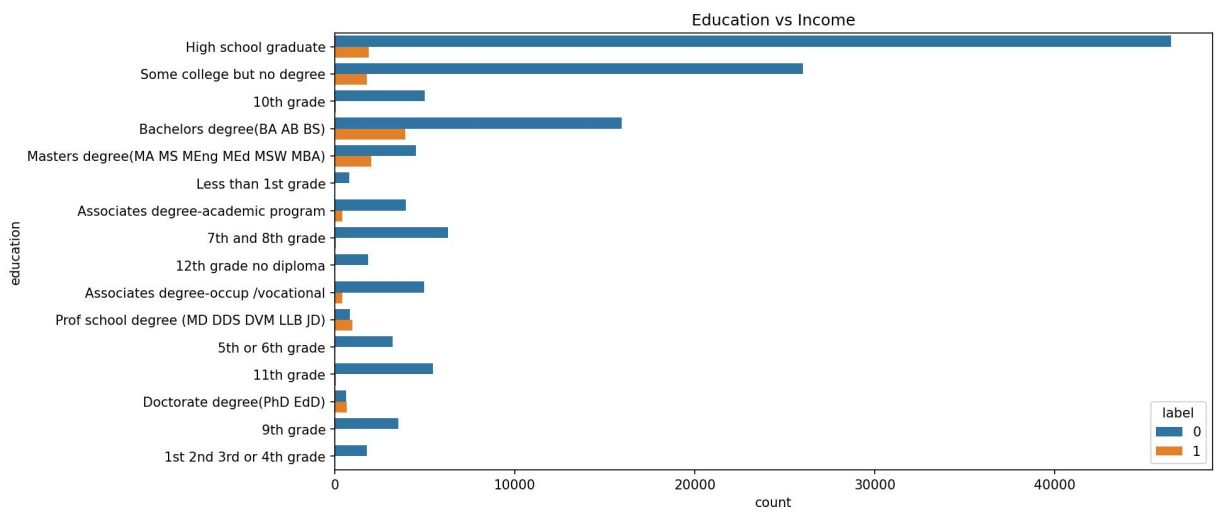

Figure 2: Education level by income group

Sex and income

Males represented approximately 75% of high earners despite females being the larger group in the dataset overall. This reflects the documented wage gap of the mid-1990s and is an important consideration for the client's targeting approach.

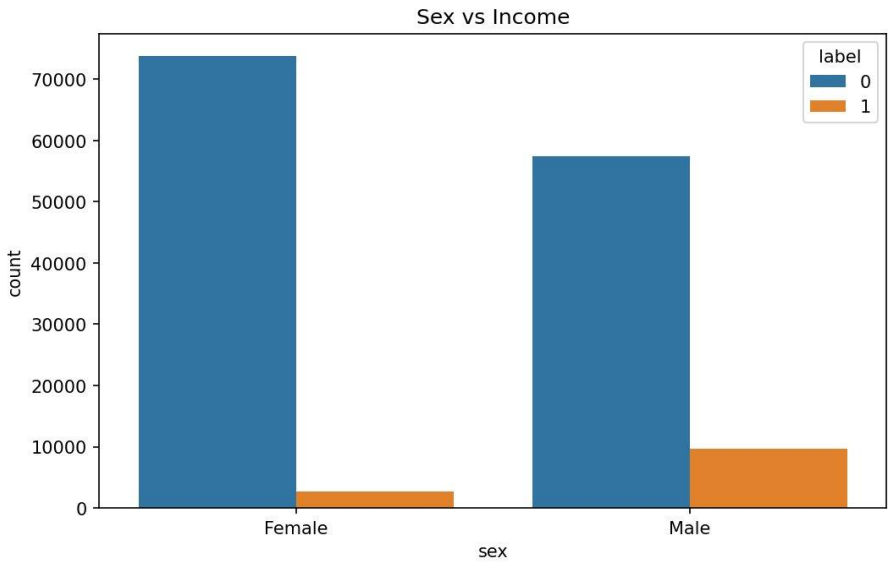

Figure 3: Sex distribution by income group

4. Data Preparation

Several cleaning and preparation steps were applied before modeling. Each decision is documented with its justification.

Step	Action	Justification
Remove minors	Removed 55,992 records for individuals under 18	Cannot legally work full time; 55,990 of 55,992 were labeled under $50K
Drop migration columns	Removed 4 migration-related features	50% missing values in all four simultaneously; not reliably imputable
Drop zero-variance column	Removed family members under 18	Single unique value remaining after age filter; no predictive power
Handle missing values	Replaced unknown values with category 'Unknown'	Preserves the information that data was unavailable without guessing
Encode label	0 = under $50K, 1 = over $50K	Required for binary classification models
Fix column types	Converted 5 numeric code columns to categorical	Values represent categories, not measurements
Train/test split	80% train, 20% test with stratified sampling	Stratification preserves class ratio in both sets

Part A: Classification Model

5. Classification Approach

5.1 Encoding Strategy

Categorical variables were encoded using target encoding, which replaces each category with the proportion of individuals in that category who earn over $50,000. This directly captures the relationship between each category and the income outcome, reduces the feature space from 450+ columns (one-hot) to 35 columns, and was found to perform on par with or better than one-hot encoding. Encoding statistics were always calculated only from training data to prevent data leakage.

5.2 Models Evaluated

Seven model types were evaluated. Each was configured to account for the class imbalance through built-in class weighting or adjusted positive class weight.

Model	Imbalance Strategy	Outcome
Logistic Regression	Balanced class weights	Baseline reference
Random Forest	Balanced class weights	Evaluated
XGBoost	Dynamic positive class weight (~10:1)	Evaluated; both encoding types tested
LightGBM	Built-in imbalance flag	Best individual model
CatBoost	Automatic class balancing	Evaluated
Neural Network (MLP)	Threshold tuning post-training	Used in final ensemble
SVM / KNN	N/A	Ruled out: computationally infeasible at this data scale

5.3 Hyperparameter Tuning

Randomized search with 3-fold cross-validation was applied to XGBoost, LightGBM, and the Neural Network. LightGBM achieved the best result with a ROC-AUC of 0.9374 after tuning.

5.4 Ensemble and Threshold Tuning

The tuned LightGBM and tuned Neural Network were combined into a soft voting ensemble. These two models were selected because they complement each other: LightGBM produces high recall while the Neural Network produces higher precision. Soft voting averages their probability outputs, producing the best F1 score of all approaches evaluated. The optimal classification threshold of 0.5382 was then identified from the precision-recall curve, improving precision from 0.56 to 0.61.

6. Classification Results

6.1 Feature Importance

After training, XGBoost's feature importance scores revealed which variables are most predictive of income. The results confirmed and extended what was found during data exploration.

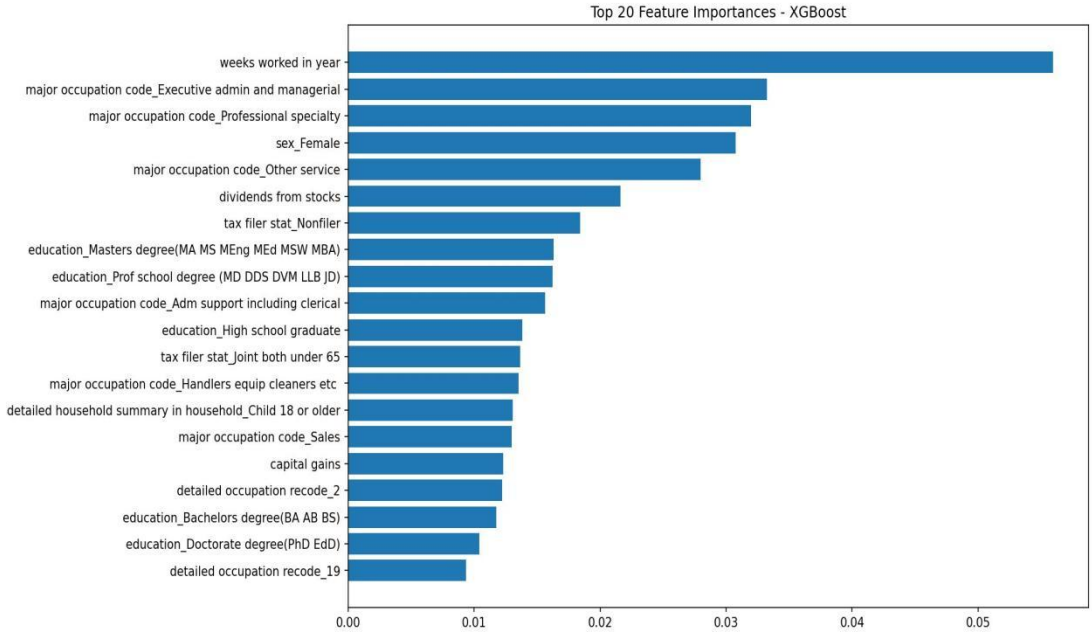

Figure 4: Top 20 features by XGBoost importance score

Weeks worked in year is the single strongest predictor by a clear margin, ranking above education and occupation. This suggests that employment consistency is a more direct indicator of annual income than job title or credentials alone. Occupation type, dividend income, and education level follow as the next most important predictors.

6.2 Model Comparison

Model	Recall	Precision	F1	ROC-AUC
LightGBM (tuned)	0.86	0.37	0.52	0.9374
XGBoost (tuned)	0.83	0.39	0.53	0.9372
CatBoost	0.86	0.36	0.50	0.9336
Ensemble + Threshold Tuning	0.64	0.61	0.62	0.9359
Ensemble LightGBM + MLP	0.68	0.56	0.62	0.9359
Logistic Regression	0.86	0.33	0.47	0.9239
Neural Network (tuned)	0.41	0.73	0.53	0.9269
Random Forest	0.40	0.74	0.52	0.9233

All metrics reported for the high-income class. ROC-AUC reflects overall model quality independent of the classification threshold.

6.3 Recommended Models

Option A: Tuned LightGBM

Of 2,476 actual high earners in the test set, LightGBM correctly identified 2,129 (Recall 0.86). It missed 347 and incorrectly flagged approximately 3,600 low earners. Best for campaigns where maximum reach is the priority.

Option B: Ensemble with Threshold Tuning

Correctly identified 1,585 of the 2,476 high earners (Recall 0.64) while generating significantly fewer false positives. For every 100 people flagged as high earners, 61 are correctly classified. Best for campaigns where marketing efficiency is equally important as reach.

Part B: Segmentation Model

7. Segmentation Approach

The segmentation model goes beyond the binary income split to identify natural customer groups within the population. Two complementary segmentations were produced: one for the full adult population and one focused exclusively on high earners.

7.1 Methodology

- Encoding: Target encoding using income label for the full population and weeks worked in year for the high earner subgroup.
- Dimensionality reduction: PCA reduced 35 features to 18 components (full population) and 19 components (high earners), each retaining 85% of the total variance. This ensures distance calculations in clustering are meaningful.
- Clustering: K-Means with optimal cluster count determined by silhouette score and elbow method. Five clusters were selected for the full population (best silhouette score 0.23 at k=5). Two clusters were selected for high earners (silhouette score 0.52 at k=2, dropping sharply to 0.16 at k=3, indicating a genuinely strong natural split).

8. Segmentation Results

8.1 Full Population Segments

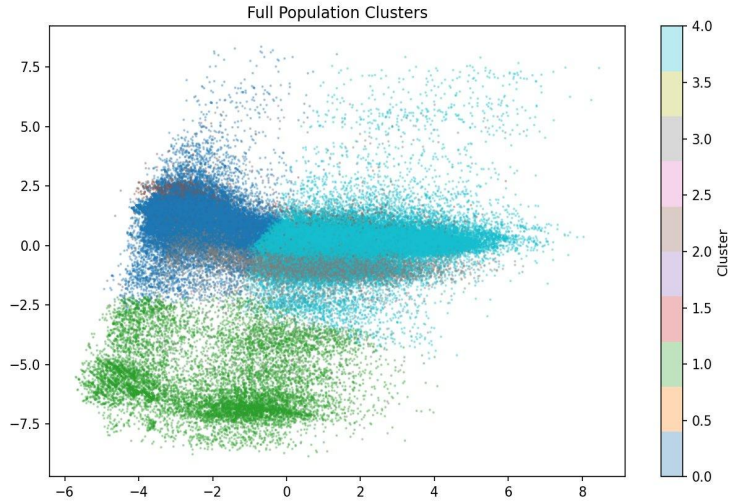

Figure 5: Full population cluster visualization (PCA components 1 & 2)

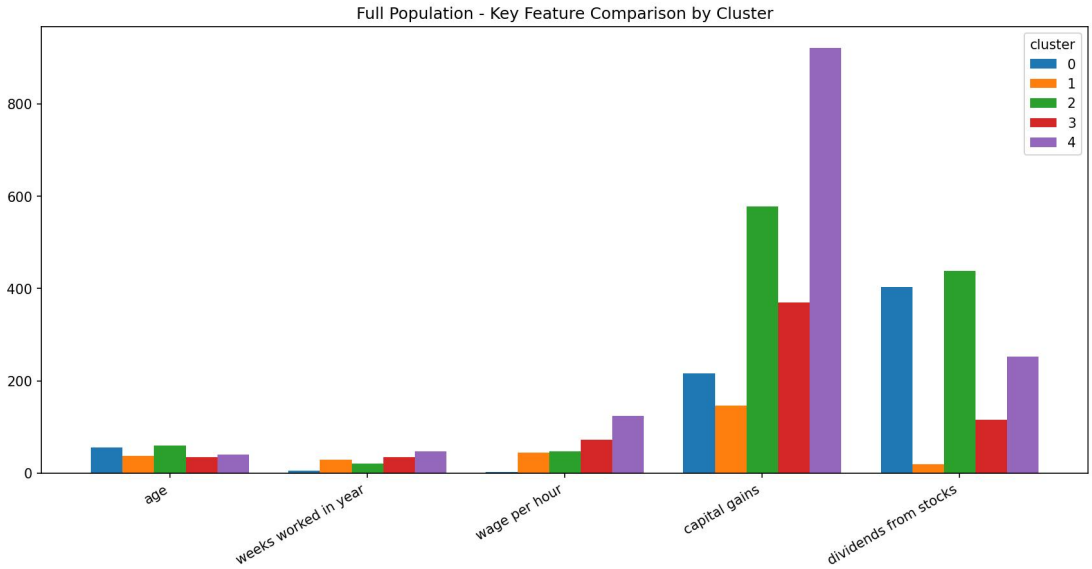

Figure 6: Key feature comparison across full population clusters

Segment	Size	Avg Age	Avg Weeks	Avg Wage/Hr	Avg Cap. Gains	Avg Dividends
Working Professionals (Cluster 4)	76,899 (53.6%)	40	48	$123	$921	$253
Retired/Inactive Homemakers (Cluster 0)	44,335 (30.9%)	56	5	$2	$216	$402
Part-time Workers (Cluster 1)	10,867 (7.6%)	38	28	$44	$146	$19
Mid-level Workers (Cluster 3)	9,839 (6.9%)	35	34	$72	$370	$116
Wealthy Inactive/Investors (Cluster 2)	1,591 (1.1%)	59	20	$48	$578	$439

Working Professionals (53.6%)

The dominant segment. Active professionals in their late 30s to mid-40s, working close to full year in professional specialty and executive roles in the private sector. Highest wages and financial activity of all segments. Primary target for premium products and investment services.

Retired & Inactive Homemakers (30.9%)

Predominantly female, married, average age 56, with minimal employment activity. Moderate investment income. Lower priority for income-based targeting but relevant for household and lifestyle products

Part-time Workers (7.6%)

Similar demographics to the inactive group but with meaningful part-time employment averaging 28 weeks. May respond to career advancement or flexible financial products.

Mid-level Workers (6.9%)

Younger at average age 35, working more weeks at a higher wage than part-timers. On a clear upward trajectory. Strong growth potential for aspirational and financial advancement products.

Wealthy Inactive / Investors (1.1%)

Small but financially distinct. Average age 59, working very few weeks, with the highest investment income of any segment outside high earners. Despite limited employment, they are high-value targets for wealth management, luxury goods, and high-end leisure products.

8.2 High Earner Segments

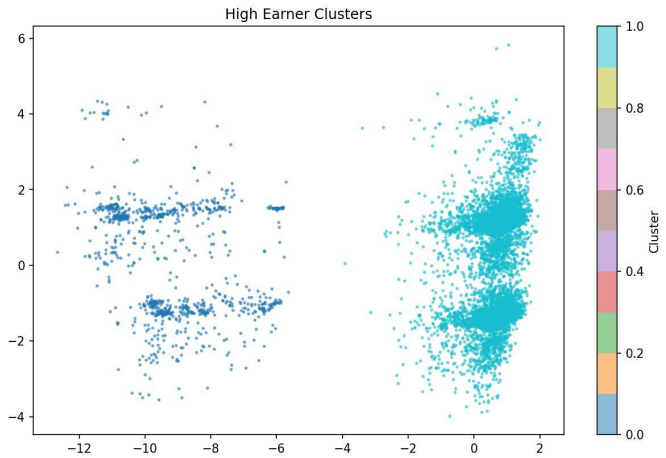

Figure 7: High earner cluster visualization (PCA components 1 & 2). Clear separation confirms two distinct groups.

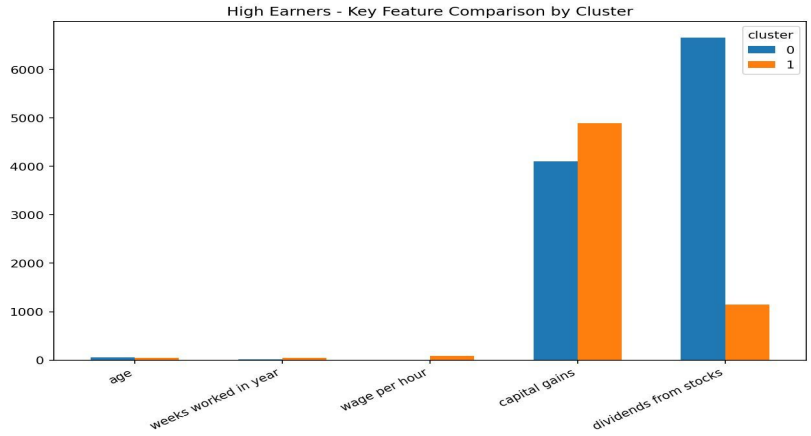

Figure 8: Key feature comparison across high earner clusters

Segment	Size	Avg Age	Avg Weeks	Avg Wage/Hr	Avg Cap. Gains	Avg Dividends
High Earning Professionals (Cluster 1)	11,474 (92.7%)	45	51	$88	$4,889	$1,151
Wealthy Investors & Retirees (Cluster 0)	906 (7.3%)	63	14	$0	$4,103	$6,657

High Earning Professionals (92.7%)

Active professionals in their mid-40s working nearly full year in executive and managerial roles. Strong wages and capital gains income. Primary premium customer base, highly receptive to career-oriented luxury products and premium financial services.

Wealthy Investors & Retirees (7.3%)

Average age 63, largely retired with zero hourly wage. Entirely investment-driven income: average dividends of $6,657 and capital gains of $4,103 make them the wealthiest segment in the entire dataset by passive income. Should be targeted with luxury lifestyle, travel, and wealth management products rather than career-related offerings.

9. Marketing Recommendations

9.1 Classification Model Deployment

We recommend deploying both classification models for different campaign types:

- New customer acquisition: Use the Ensemble with Threshold Tuning. When marketing to unfamiliar individuals, efficiency matters. This model significantly reduces spend on contacts unlikely to convert.
- Upselling to existing customers: Use the Tuned LightGBM. When the cost of missing a high-value customer outweighs the cost of extra contacts, the higher recall justifies the lower precision.

9.2 Segmentation-Based Strategy

Segment	Priority	Recommended Approach
Working Professionals	High	Premium products, financial services, career tools, investment platforms
Mid-level Workers	High	Aspirational products, career advancement, savings and investment tools
Wealthy Inactive/Investors	Medium-High	Luxury goods, wealth management, leisure and lifestyle products
Part-time Workers	Medium	Value-driven premium offers, flexible financial products
Retired/Inactive Homemakers	Low	Household and lifestyle products; not income-targeted campaigns
High Earning Professionals	Very High	Career luxury, executive travel, premium financial services
Wealthy Investors & Retirees	Very High	Luxury lifestyle, travel, estate planning, premium leisure

Treating both high earner segments identically would be a significant missed opportunity. The same luxury product positioned differently for each segment will achieve substantially better conversion rates.

9.3 Key Customer Profile Insights

Across both the classification and segmentation models, several consistent findings stand out:

- Employment consistency is the strongest income predictor. Individuals working close to 52 weeks per year are far more likely to earn over $50,000 than those with gaps or part-time arrangements.
- Occupation drives income more than industry. Executive, managerial, and professional roles are associated with high income regardless of which sector the person works in.
- Investment income signals wealth. Individuals with dividend or capital gains income are highly likely to also have a high salary.
- Education is a near-prerequisite but not a guarantee. Almost all high earners hold at least a Bachelor's degree, but many degree holders still fall below the threshold.

9.4 Important Considerations

This model was built on data from 1994 and 1995. Income levels, workforce participation, and the relationship between demographic variables and earnings have shifted considerably over the past 30 years. We strongly recommend retraining on more recent data before deploying in a live marketing context.

Sex is among the higher-importance features in the classification model, reflecting the wage gap documented in the mid-1990s. Relying on this variable for individual-level targeting decisions may raise ethical and legal concerns and warrants review by the client's compliance team.

10. Limitations and Future Work

- Data recency: The 1994-1995 census data limits real-world applicability. Retraining on contemporary data is the highest-priority next step.
- Population weights: The dataset includes sampling weights that were not incorporated into model training. Using them could improve how well the model reflects the true population distribution.
- Individual-level explanations: The current model produces a score per individual but does not explain why. Implementing SHAP-based explanations would allow the client to understand the specific factors driving each prediction.
- Campaign-specific thresholds: The current threshold maximizes overall F1. For campaigns where the client can quantify the cost of a missed high earner vs. a wasted contact, a custom threshold from those cost parameters would be more financially optimal.
- Segmentation stability: K-Means cluster assignments can shift with different random seeds or data samples. Cross-validating cluster stability before acting on the segment definitions is recommended.
- Alternative clustering methods: K-Means assumes spherical clusters of similar size. The large size difference between segments suggests some groups may benefit from DBSCAN or Gaussian Mixture Models in future work.
- Behavioral data: Census data captures demographics but not purchasing behavior or brand affinity. Combining these segments with actual purchase history would significantly improve targeting precision.